Advanced information, computation, communication I
EPFL - Fall semester 2020-2021

Week 10

November 17, 2020

Exercise 1.

You are in the city of La Chaux-de-Fonds, globally famous for the orthogonality of its streets.

You are currently located at the intersection of Rue Numa-Droz with Rue du Maire-Sandoz (START), and your goal is to arrive to the intersection of Rue de la Serre with Rue du Balancier (END) where you parked your bike. You can take all the streets, no matter how narrow (for example Rue de la Paix is usable, as well as Rue de l'Avenir (name not shown on map - one where CIFOM Centre is located)), but you cannot take the closed roads (there is one closed road on the map).

Due to a recent referendum in the canton of Neuchâtel that passed with a tight majority (50.7 % - 49.3 %), all EPFL students can only walk to the right or down when moving by foot in the city of La Chaux-de-Fonds. Since you left your bike at (END), your only solution is to go by foot, and you need to obey the current laws. Naturally, this new constraint on moving sparked an interest in you, and you decided to calculate in how many ways can one walk from (START) to (END) following the rules of only being able to move to the right or down. In other words, how many different paths exist from (START) to (END) without backtracking?

NB. You do not need to go to La Chaux-de-Fonds in order to solve this exercise.

Exercise 2. Each user on a computer system has a password, which is seven or eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least two digits. How many possible passwords are there?

Exercise 3.

1. A group contains n men and n women. How many ways are there to arrange these people in a row if the men and women alternate?

2. Suppose that a department contains 10 men and 15 women. How many ways are there to form a committee with six members if it must have the same number of men and women?

Exercise 4. An exam has 12 questions, with 4 possible answers for each question. How many students should complete the exam to ensure that at least 3 students will submit the exact same answers?

Exercise 5.

1. How many functions are there from $A = \{0, 1, 2, 3\}$ to $B = \{0, 1, 2\}$?
2. How many injective functions are there from $A = \{0, 1, 2, 3\}$ to $B = \{0, 1, 2, 3, 4, 5, 6\}$?

Exercise 6. Suppose that in the future every telephone in the world is assigned a number that contains a country code that is 1 to 3 digits long that is of the form X, XX, XXX followed by a 10-digit telephone number of the form NXX-NXX-XXXX, where N can take any values from 2 through 9 and X any values from 0 to 9. How many unique phone numbers would be available worldwide according to this numbering plan?

- ◯ 12876000
- ◯ 7.104×10^{12}
- ◯ 6.4×10^{15}
- ◯ 3.058×10^{12}

Exercise 7. How many bit strings of length 10 contain

1. exactly four 1s?
2. at most four 1s?
3. at least four 1s?
4. an equal number of 0s and 1s?

Exercise 8.

1. The number of distinct triples (x_1, x_2, x_3) of non-negative integers x_1, x_2, x_3 such that $x_1+x_2+x_3 = 8$ equals
   - ◯ 495.
   - ◯ 330.
   - ◯ 165.
   - ◯ 45.
2. The number of distinct triples (x_1, x_2, x_3) of non-negative integers x_1, x_2, x_3 such that $x_1+x_2+x_3 \leq 8$ equals
   - ◯ 495.
   - ◯ 330.
   - ◯ 165.
   - ◯ 45.
3. The number of distinct quadruples (x_1, x_2, x_3, x_4) of non-negative integers x_1, x_2, x_3, x_4 such that $x_1 + x_2 + x_3 + x_4 < 8$ equals
   - ◯ 495.
   - ◯ 330.
   - ◯ 165.
   - ◯ 45.

4. The number of distinct quadruples (x_1, x_2, x_3, x_4) of non-negative integers x_1, x_2, x_3, x_4 such that $x_i \geq i$ and $x_1 + x_2 + x_3 + x_4 \leq 18$ equals

   ◯ 495.
   ◯ 330.
   ◯ 165.
   ◯ 45.

5. The number of distinct quintuples $(x_1, x_2, x_3, x_4, x_5)$ of non-negative integers x_1, x_2, x_3, x_4, x_5 such that $x_1 \geq 3$, $x_2 \geq 3$, $x_3 \geq 0$, $x_4 \geq 8$ and $0 \leq x_5 \leq 3$, and $x_1 + x_2 + x_3 + x_4 + x_5 < 24$ equals

   ◯ 2002.
   ◯ 1750.
   ◯ 715.
   ◯ 210.

Exercise 9. How many distinct five-card poker hands contain:

1. *One pair* (poker hand containing two cards of the same kind and three cards of three other kinds).
2. *Two pairs* (poker hand containing two cards of the same kind, two cards of another kind and one card of a third kind).
3. *Three of a kind* (poker hand containing three cards of the same kind and two cards of two other kinds).

Exercise 10. Find the least number of cables required to connect 15 computers to 10 printers to guarantee that any group of 10 computers can always directly access 10 different printers simultaneously:

◯ 60

◯ 150

◯ 10

◯ None of the above

Exercise 11. Given $n \in \mathbf{N}$, let $\mathbf{v}_1, \ldots, \mathbf{v}_s$ be a collection of s n-dimensional vectors of integers (i.e., $\mathbf{v}_i \in \mathbf{Z}^n$ for $1 \leq i \leq s$). How large should $s \in \mathbf{N}$ be such that at least one of the middle points (i.e., the vector $\frac{\mathbf{v}_i + \mathbf{v}_j}{2}$ for any two indices $1 \leq i < j \leq s$) is an all-integer vector?

Exercise 12. Prove the hockey-stick identity using a mathematical argument (as opposed to a combinatorial argument):

For any integers n and r with $0 \leq r \leq n$, we have

$$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}.$$